

Artificial Intelligence (AI) has revolutionized how humans interact with technology. Large Language Models (LLMs)

However, LLMs suffer from two major limitations:

1. Knowledge Cutoff: An LLM only knows information up to its training date.
2. Hallucinations: When an LLM does not know a specific fact, it may fabricate a plausible-sounding answer.

To solve this, Retrieval-Augmented Generation (RAG) was introduced. RAG connects an LLM to external private

An Operating System (OS) manages hardware resources and provides common services for programs. The core

Process States:

- Ready: The process is waiting to be assigned to a CPU core.
- Running: Instructions are actively being executed by the processor.
- Waiting/Blocked: The process is waiting for an I/O event or signal.

Virtual Memory is a memory management technique that provides an idealized abstraction of storage. It uses Pa

All engineers must adhere to strict API security guidelines when deploying cloud services. API keys, database credentials, and other sensitive information must be handled with care.

Key Security Rules:

1. Store all secrets inside environment variables and load them via .env files.
2. All client communication must occur over HTTPS with TLS 1.3 encryption.
3. Rate limiting is set to a maximum of 60 requests per minute per IP address.
4. In case of a credential leak, the security response team must revoke the key within 15 minutes.